Categories for the Working Mathematician - Self Study

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Abstract

This text contains my notes on Categories for the Working Mathematician, by Saunders Mac Lane. I read this book independently, starting in mid spring semester of my sophomore year, 2025. My notes largely paraphrase the text, and I work out whatever examples I see fit.

Contents

Chapter 1

Categories, Functors, and Natural Transformations

1.1 Axioms for Categories

We start by describing categories using *metagraphs*, which do not employ the use of set theory. A metagraph consists of objects, arrows, and two operatons:

- Domain, which maps arrows to an object $\operatorname{dom} f = a$;
- Codomain, which maps arrows to an object $\operatorname{cod} f = b$.

We write $f : a \to b$ to indicate f is an arrow with $\operatorname{dom} f = a$ and $\operatorname{cod} f = b$.

A *metacategory* is a metagraph together with two additional operations:

- Identity, assigning each item a an arrow 1_a;
- Composite, assigning each pair (f, g) of arrows with $\operatorname{dom} g = \operatorname{cod} f$, an arrow $g \circ f : \operatorname{dom} f \to \operatorname{cod} g$ called their composite;

such that composition is associative and the identity arrows 1_a are identities with respect to composition. Alternative notations for the identity 1_a include id_a, and simply a itself. Examples of metacategories include the usual category **Set** of sets and **Grp** of groups, topological spaces with continuous maps as arrows, and compact Hausdorff spaces with the same as arrows.

Since identity arrows are uniquely determined for each object, it's possible to dispense entirely of the objects in a metacategory and define an arrows-only metacategory C. There exist a collection of arrows, for which some pairs (f, g) of arrows are called *composable*. For any composable arrows, we say the composition $g \circ f$ is defined, and we can then define an identity arrow to be any arrow u satisfying $f \circ u = f$ and $u \circ g = u$ for any f, g such that the composition is defined. The data must then satisfy the axioms

- $(k \circ g) \circ f$ is defined iff $k \circ (g \circ f)$ is defined, and they are equal;
- The composite kgf is defined if and only if the composites kg and gf are defined;
- For each arrow g of C, there exist identity arrows u, u' such that gu and $u'g$ are defined.

The final axiom indeed shows that identity arrows are unique in this regard, and allows defining of a domain and codomain for the arrow g. An arrows-only metacategory thus satisfies the usual metacategory axioms, if identity arrows are taken as objects.

1.2 Categories

Definition 1.1: Directed Graph

A directed graph is a set O of objects, a set A of arrows, and two functions

$$A \underset{\text{cod}}{\overset{\text{dom}}{\rightrightarrows}} O$$

where the set of composable pairs is the set

$$A \times_O A = \{(g, f) \mid g, f \in A, \operatorname{dom} g = \operatorname{cod} f\}.$$

Definition 1.2: Category

A category is a directed graph with two additional functions

$$\begin{array}{ccc} O \xrightarrow{\text{id}} A, & \qquad & A \times_O A \xrightarrow{\circ} A, \\ \\ c \longmapsto 1_c, & \qquad & (g, f) \longmapsto g \circ f, \end{array}$$

where $g \circ f$ is usually written gf, such that

$$\operatorname{dom} 1_a = \operatorname{cod} 1_a, \qquad \operatorname{dom} gf = \operatorname{dom} f, \qquad \operatorname{cod} gf = \operatorname{cod} g.$$

for all $a \in O$ and $(g, f) \in A \times_O A$, such that a metacategory is formed.

That is, a category is an interpretation of a metacategory within set theory. If C is a category, we usually write $a \in C$ or $f \in C$ to say a is an object of C, or f is an arrow of C. We also write

$$\operatorname{Hom}(b, c) = \{f \in C \mid \operatorname{dom} f = b, \operatorname{cod} f = c\}$$

for the set of arrows $b \to c$. We will show later that this definition of a category is the same as saying a category is a monoid for the product $\times_O$.

Examples of categories include

- $\mathbf{0}$, the empty category;
- $\mathbf{1}$, the category with one object and one arrow;
- $\mathbf{2}$, the category with two objects and precisely one non-identity arrow $a \to b$;
- $\mathbf{3}$, the category with three objects whose non-identity arrows form a triangle

$$\begin{tikzcd}
& \bullet \arrow[dr] & \\
\bullet \arrow[ur] & & \bullet \arrow[ll]
\end{tikzcd}$$

- $\downdownarrows$, the category with two objects and two non-identity arrows $a \rightrightarrows b$; we call such arrows *parallel arrows.*

A category is called *discrete* when all arrows are identities. Any set forms a discrete category, and any discrete category is completely determined by it's arrows.

A *monoid* is a category with one object. A monoid is thus fully described by it's arrows, and it's composition operation. A monoid can thus be described as a set M (of arrows) together with a binary operation $M \times M \to M$ and an identity element; such an object is precisely a semigroup with unity. Thus, for any category C and any object $a \in C$, the category $\mathrm{Hom}(a, a)$ forms a monoid. In this vein of thought, a group is thus a category where every arrow has a two-sided inverse.

Given a set V of sets, the category $\mathbf{Ens}_V$ takes V as the objects, and morphisms as any regular functions between sets in V. **Ens** refers to an arbitrary category of this type.

An ordinal number n can be seen as the set $\{0, \ldots, n-1\}$, and hence each ordinal is a category when viewed as a preorder. We can also consider the category ω, the first infinite cardinal, which when viewed as a category looks like

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \ldots$$

The category Δ takes all finite ordinals as objects and all order-preserving functions $f : m \to n$ as arrows. This category is generally called the *simplical category*, and it is extremely important.

We want an actual category **Set**, which is clearly not a set itself. We will circumvent this by allowing there to exist a "universe" $\mathcal{U}$, and will call a set "small" if it belongs to the universe. We will construct this explicitly later (and I will be calling small sets "sets" and large sets "classes" like a normal fucking person). This allows us to construct

- **Set**, objects are sets and morphisms are functions;
- $\mathbf{Set}_*$, objects are sets with a selected point, and morphisms are functions preserving these points;
- **Ens**, category of all classes and functions within a variable set V;
- **Cat**, category of small categories and morphisms as functors;
- **Grp**, category of groups and morphisms as group homomorphisms;
- **Ab**, category of abelian groups;
- **Mon**, category of (small) monoids;
- **Ring**, category of rings;
- **CRing**, category of commutative rings;
- R-**Mod**, category of left R-modules;
- **Mod**-R, category of right R-modules;
- **Top**, category of topological spaces;
- **Toph**, category of topological spaces with morphisms as homotopy classes of maps;
- $\mathbf{Top}_*$, category of topological spaces with a distinguised base point; base preserving continuous maps.

1.3 Functors

Definition 1.3: Functor

Let C, D be categories. A functor $T : C \to D$ consists of an object function T mapping each $c \in C$ to some $Tc \in D$, and an arrow function mapping each $f : c \to c'$ in C to $Tf : Tc \to Tc'$ in D, in such a way that

$$T1_c = 1_{Tc}, \qquad T(g \circ f) = T(g) \circ T(f)$$

whenever the composite $g \circ f$ is defined in C.

For example, the power set functor $\mathscr{P} : \mathbf{Set} \to \mathbf{Set}$, defined such that for any $f : A \to B$, the new map $\mathscr{P}f : \mathscr{P}A \to \mathscr{P}B$ is such that for any $S \subseteq A$, $\mathscr{P}f(S) = f(S)$.

Functors first arose in algebraic topology. For any n, a topological space X can be assigned an abelian group $H_n(X)$ called the n-th homology group of X. However, for any $: X \to Y$, there exists an induced $H_n(f) : H_n(X) \to H_n(Y)$ morphism of groups, and thus H_n becomes a functor $\mathbf{Top} \to \mathbf{Ab}$. We can in fact show homologygroups are invariant under homotopy, and so we can equally define a functor $\mathbf{Toph} \to \mathbf{Grp}$. The homotopy groups π_n are also functors; since they depend on a base point they define a functor $\mathbf{Top}_* \to \mathbf{Grp}$.

A functor which "forgets" underlying structure of an object is called a *forgetful functor*. For example, there is an apparent forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$, another $U : \mathbf{Ring} \to \mathbf{Ab}$, and so on.

Clearly functors may be composed, and moreover there is an identity functor 1_C for any category C. We may thus consider the metacategory of all categories, or the category $\mathbf{Cat}$ of all small categories.

A functor is called an *isomorphism* if it is two-sided invertible; in general isomorphism is too strong a condition, and weaker conditions are more useful. A functor is *full* if for every $c, c' \in C$ and every arrow $g : Tc \to Tc'$ in D, there exists $f : c \to c'$ in C such that $g = Tf$. A functor is *faithful*, or an embedding, if for all $c_1, c_2 \in C$ and parallel arrows $f_1, f_2 : c_1 \to c_2$ such that $Tf_1 = Tf_2$, we have $f_1 = f_2$.

These properties can be realized more intuitively. A functor defines functions $T : \mathrm{Hom}(c, c') \to \mathrm{Hom}(Tc, Tc')$. A functor is full when these functions are surjective; it is faithful when they are injective. A fully faithful functor is then a functor where these functions are bijections.

A subcategory S of a category C is a subset of the objects and arrows in C in such a way that S forms a category. There is automatically a faithful inclusion functor $S \hookrightarrow C$; we say a subcategory is *full* when the inclusion functor is full. Intuitively, S contains all arrows in C for the specified subcollection of objects.

1.4 Natural Transformations

Definition 1.4: Natural Transformation

Let $T, S : C \to D$ be functors. A natural transformation $\tau : S \Rightarrow T$ is a function which assigns to each $c \in C$ an arrow $\tau_c = \tau c : Sc \to Tc$ of D, called the *component of* τ *at* c, such that for all $f : c \to c'$ in C, the diagram

$$\begin{tikzcd} Sc \arrow[r, "\tau c"] \arrow[d, "Sf"'] & Tc \arrow[d, "Tf"] \\ Sc' \arrow[r, "\tau c'"] & Tc' \end{tikzcd}$$

commutes.

We say any component $\tau c : Sc \to Tc$ is *natural* in c. We can think of a natural transformation $\tau : S \Rightarrow T$ as a way of transforming the image of a functor S into that of a functor T. For example, given a triangle

$$\begin{tikzcd} a \arrow[dr, "f"] \arrow[dd, "h"'] & \\ & b \arrow[dl, "g"] \\ c & \end{tikzcd}$$

we would expect the diagram

$$\begin{tikzcd}
Sa \arrow[rr, "\tau a"] \arrow[dr, "Sf"] \arrow[dd, "Sh"'] & & Ta \arrow[dr, "Tf"] \arrow[dd] & \\
& Sb \arrow[rr] \arrow[dl, "Sg"] & & Tb \arrow[dl, "Tg"] \\
Sc \arrow[rr, "\tau c"'] & & Tc &
\end{tikzcd}$$

commute. Indeed, it does.

Natural transformations define *morphisms of functors*. If a natural transformation has every component invertible, we call it a *natural isomorphism* and write $S \cong T$.

For example, the determinant is a natural transformation $\det_K : \mathrm{GL}_n \Rightarrow ()^*$, where $K \mapsto K^*$ maps a ring to it's group of units. These are clearly two functors